Violence Detection Using Affective Features from Upper Body Expressions

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In recent years, places such as elderly care houses, nursing homes, schools, prisons, psychic wards are prone to aggressive and violent and abusive interactions. This leads to the need for a system that could automatically detect such abhorrent behaviours. With the rapid increase in the use of surveillance cameras, the implementation of such a system can help in deterring such intolerable incidents and thus providing a much safer environment. In the paper, a violence detection system is proposed using affective features for the anger emotion in order to enhance to efficiency of the existing violence detection systems. OpenPose toolbox is used for extracting 2D skeleton keypoints from the RWF2000 dataset and MATLAB Classification Learners app is used for classification, producing encouraging results.

CCS Concepts: • **Computing methodologies** → **Scene anomaly detection**;

KEYWORDS: Violence detection, OpenPose, 2D skeleton, affect recognition, RWF2000

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Computer vision-based Human action recognition (HAR) is the ability of computers to recognize human actions based on the given video input [1]. For decades, many researchers have developed algorithms to tackle action recognition problems ranging from human tracking, activity recognition, crowd monitoring, smart surveillance and many more. This has resulted in a vast understanding of the field, as with time, researchers have started to narrow their focus on more specific areas. Lately, hostile, violent, aggressive action recognition or fight detection has gained increasing popularity among researchers. With applications such as surveillance and safer internet, there have been many proposed algorithms to detect violent activities [2], [3]. When it comes to the problem of abuse detection, actions such as punching, slapping, kicking, pushing and poking could be identified with the help of hostile action recognition algorithms and research. However, the current real-time surveillance systems focus on violence, fight or hostile action detection. These systems emphasize on two-way fights involving two or more people and comprising of huge and vigorous movements rather than abusive behaviour subjected to someone [3], [4]. To enhance the efficiency of the existing surveillance systems, the emotion content should also be taken into account [5].

Detection and recognition of a variety of emotions is being studied under the umbrella of affective computing, ranging from expressions such as anger, surprise, happiness, sadness, anxiety, and disgust [6], [7]. Since the inception of the term and concept of affective computing, many researchers have developed automatic affect detection systems that are able to detect emotions from different body features. However, majority of the work is based on facial expressions [8], until fairly recently, researchers have started to take interest in bodily or mixed features. As violent and abusive behaviour is particularly demonstrated through the body, aggressive and anger is notably expressed through the arms [9, 10, 11]. In this paper, the recognition of violence detection is performed using low-level features by mapping behavioral cues defined by psychologists from the upper body.

In this section, the existing hostile action recognition and violence detection research is covered. A comprehensive review of the techniques implementing machine learning and deep learning frameworks is along with the analysis and comparison of the state-of-the-art approaches is discussed. Figure 1 shows the organizational chart of the violence detection literature covered, also highlighting the direction towards the proposed methodology.

Figure 1

Figure 1: Structure of the literature reviewed for violence detection. The orange box indicates the area covered in the proposed methodology

Traditional methods of action recognition, be it violent or non-violent are majorly an amalgamation of two processes; handcrafted features extraction and machine learning techniques [12].

Extracting handcrafted features require expertise and advance knowledge and mainly comprise of motion based, trajectory based, appearance based etc.

2.2.1 Motion based. Motion based features provide temporal information through varying frames [12], [13]. The earlier works on action recognition using motion-based features are Motion Energy Image (MEI) and Motion History Image (MHI) [14]. MEI and MHI extract global features which represent wholesome information of the video [15]. There is some notable work in the area of hostile action recognition using motion features since the last two decades. [16] used local binary motion descriptors and SVM as a classifier to detect aggressive behaviour using the surveillance-based CareMedia dataset. In [17], a violent detection system for trains using MHI and energy signatures was proposed applying the Mean-shift algorithm to compute the similarity between events. In [18], a new descriptor namely Violent flows (ViF) was introduced by considering how flow-vector magnitudes change over time, and classification was performed using the SVM classifier. [19] proposed an Oriented Histogram of Optical Flow (OHOF) descriptor and used the SVM classifier to detect violent scenes in surveillance footage. In [20], a fast fight detection method was proposed where violent interactions were detected using blobs analysis. The results were much faster than the state-of-art technologies but at the expense of lower accuracy. [21] proposed a new violent detection descriptor using an aggregated histogram of Weber Local Descriptors (WLD) for spatial information, and then an aggregated HOF was computed for motion information. Sparse coding was used to convert the low-level descriptors to mid-level features and SVM was used as a classifier. In [22], the authors have proposed a violent detection method using oriented Violent Flows whereby not only the motion magnitude but also the motion orientation was taken into account. Many researchers also combine motion information with other types of features for higher efficiency, such as appearance-based features.

2.2.2 Appearance based. Appearance based features are also commonly used along with motion features, these methods take into account the variations in appearance. Appearance based features on their own may not be discriminative enough for violence and hostile behaviour detection, hence are often merged with other types of techniques. The work in [23] using Space-Time Interest Points (STIP) proved very crucial in this domain. In their seminal paper, the authors used STIP (HOG and HOF), MoSIFT, and BoW to extract features and used SVM as the classifier. Two new datasets were also introduced, namely Hockey and Movies. These datasets have become standard datasets used for violent detection [3], [24]. [25] proposed a violent detection method by combining MoSIFT with Motion Boundary Histogram (MBH) and movement filtering algorithm.

2.2.3 Trajectory based. In [26], the authors proposed a spatio-temporal elastic cuboid trajectory with Hough forests. The network was tested on UT-1, UT-2, Movies, Hockey datasets. In [27] violence detection method for indoor surveillance cameras using both global and local features was presented. The authors used Kalman filter and standard deviation to calculate the global motion trajectory and dense sampling to extract STIP volumes following optical flow using Lucas-Kanade. An SVM classifier was used for classification on the CAVIAR dataset.

2.2.4 Other methods. The authors in [2] proposed a Spatio-temporal improved fisher vector (IFV) for violent detection. In [28], three different types of features were merged together to detect hostile behaviour at a counter. These features were the Kinect skeletal 3D data, Posture and gesture frequency features, kinetic motion and RGB-D depth sensor for silhouette.

Machine learning develops learning algorithm from sample data to generate models that can make predictions on new data [29]. Handcrafted features are then fed to machine learning models for classification purposes. Most commonly machine learning algorithms are classified as supervised learning, semi-supervised learning and un-supervised learning. In supervised learning the labelled training data is fed to the classifier and the model learns the classes and classifies the unknown test data into given classes. **Support Vector Machine** (SVM) is the most commonly used supervised ML algorithm in the case of violence detection. In [19], [22], [30], [31], the authors used SVM along with handcrafted features for classification of hostile actions. For violence detection unsupervised learning is also applied, where training data is fed to the classifier without any labels and the model groups the inputs in different classes based on their features. **K-means clustering** is frequently used as an unsupervised classifier for violence detection problem, as in [25].

Over the years, handcrafted features have remained the backbone of hostile action recognition. However, lately, deep learning frameworks such as CNN and LSTM have gained popularity among researchers. Deep learning has gained a lot of prominence in recent years. Many researchers are proposing deep learning-based solutions for action recognition and so is the case with hostile action recognition. One of the earliest works on hostile action recognition was [32], where the authors developed a nine-layer 3D CNN trained using a stochastic gradient. In [33], the authors proposed a violent video detection system using a convLSTM network. [34] proposed a Deep Multiple Instance Learning system for anomaly detection in surveillance videos, the authors had also proposed a new dataset called UCF Crime. In [24], the authors proposed a method for violent video detection based on 3CD. A new sampling technique was introduced and applied to videos of longer length. Results were obtained on the movies, hockey and violence datasets. [35] used 3D convolution architecture (C3D) pretained on sports-1M dataset along with SVM classifier to detect violent actions. In [36] the authors used pretained models of ResNet50 along with LSTM to produce violence detection results comparable to the state-of-art.

Hybrid deep-learning based methods mostly combine deep-learning modules with handcrafted components and machine learning modules [12]. Hybrid approaches in many cases use deep-learning as a feature extractor and machine learning classifiers or handcrafted features as input to deep-learning models. [37] proposed a multi-stream deep network for person-to-person violent detection. The network computed three streams; spatial, temporal and acceleration using a 2D CNN VGG19 model pre-trained on ImageNet, ConvNet TDD model and 2D CNN respectively, and score fusion was applied. In [38], a violent detection method called FightNet based on the temporal segment network (TSN) was proposed, and three different modalities of input; RGB, optical flow and acceleration were used. In [3], the authors proposed a fight recognition system in video using Hough Forest and 2D CNNs. Recently, [39] used the hybrid approach for violence detection by computing handcrafted features and 3-stream CNN network for classification.

In this paper, the hybrid approach is considered for violence detection. Firstly, the OpenPose toolbox [40] is used for skeletal keypoints extraction from the RWF2000 dataset. The OpenPose is a deep architecture consisting of two branches with multi-stage CNNs. By extracting the 2D skeletal joint locations from RGB videos, many minuscule body movements can be detected and incorporated while detecting abusive behaviour. Using the 2D skeletal keypoints a set of features are proposed as discussed below.

As reviewed in [9], [41], many researchers and psychologist suggest the association of certain arm and upper body movements to the emotion of anger, such as arm bend, distance from a centroid, motion through frames, shoulder elevation etc. Based on those behaviour variables a set of features is proposed.

3.1.1 Keypoint location features. In figure 2(a), the 25 2D keypoints extracted through the OpenPose toolbox is shown. For the first set of features, in order to assess the movement of the arms the coordinate location of the keypoints from the wrist and elbow are considered [28], [42]. As shown in figure 2(b), a total of four keypoints are taken into account, two from the right arm and two from the left arm. These features are a frame-by-frame depiction of the movement of the arm.

Figure 2

Figure 2: (a) 2D skeleton keypoints [40] (b) Elbow and wrist keypoints (c) Elbow and shoulder angles.

3.1.2 Keypoint angles features. To estimate the amount of movement taking place during anger expression from the arms, manifestations like arm bend and arm straight or stretched forward need to be computed. These expressions can be evaluated by using trigonometric equations for computing angles. As shown in figure 2(c), the angles from the elbow and shoulder are computed for both arms. This yields a feature set of four features. The angles are calculating by using equation 1 and 2.

$$\angle elbow = \cos^{-1}\left\{\frac{(s\_e^2 + e\_w^2 - s\_w^{2)}}{(2 * s\_e * s\_w)}\right\}$$

(1)
e_s=distance between elbow and shoulder, e_w=distance between elbow and wrist, s_w=distance between shoulder and wrist

$$\angle shoulder = \cos^{-1}\left\{\frac{(h\_s^2 + s\_e^2 - h_e^2)}{(2 * h\_s * s\_e)}\right\}$$

(2)
h_s=distance between mid-hip and shoulder, s_e=distance between shoulder and elbow, h_e=distance between mid-hip and elbow

3.1.3 Temporal patches features. In order to monitor the movement of the arm joints over a window of time, temporal patches of 20 frames are proposed (Wang et al., 2013). These features encapsulate the progression of the joints over time hence assisting in recognizing an abusive action. Hence, the Euclidean distance of the present and past frame is computed for a particular keypoint as shown in equations 3 and 4. This value is a measure of the change in arm keypoints with respect to time. Using the distances computed in equations 3 and 4, the features are defined by the equations 5 and 6.

$$d_t\,(elbow) = \sqrt{\left(e_{2,t} - e_{2,(t-1)}\right)^2 + \left(e_{1,t} - e_{1,(t-1)}\right)^2}$$

(3)

$$d_t\,(wrist) = \sqrt{\left(w_{2,t} - w_{2(t-1)}\right)^2 + \left(w_{1,t} - w_{1,(t-1)}\right)^2}$$

(4)
(e_2,e_1) = elbow coordinates, (w_2,w_1) = wrist coordinates, t = current frame, t-1 = past frame

$$TP_e = \sum_{t=t-20}^{t} d_t\,(elbow)$$

(5)

$$TP_w = \sum_{t=t-20}^{t} d_t\,(wrist)$$

(6)
TPe=Temporal patches for elbow keypoint, TPs=Temporal patches for shoulder keypoint, d_t =Euclidean distance two frames for a keypoint

For this research the RWF2000 dataset is used [43]. The set comprises of 2000 videos, 1000 videos contain violence and 1000 videos are non-violent. The dataset videos are split as 80% training and 20% testing videos. The dataset is collected from YouTube, showcasing real life violence scenes. Each video contains 150 frames. Figure 3 shows a screenshot of clip from the dataset and figure 4 shows its corresponding extracted 2D skeleton. As it can be seen in figure 3, the videos are taken from surveillance cameras, which results in low resolution and occluded videos. In figure 4 it can be seen that the 2D skeleton of the person on the right is not precise enough, this makes it challenging to accurately extract the proposed features.

Figure 3
Figure 3: Screenshot from RWF2000 dataset

Figure 4
Figure 4: 2D skeleton extracted from figure 3 using OpenPose

The experiments for this paper consist of a three-part process. The 2D skeleton was extracted with the help of the OpenPose toolbox built in Python using NVIDIA RTX2060 GPU. The feature extraction was performed using MATLAB and for classification purposes, the MATLABs Classification Learners app was used. Multiple classifiers were trained using each set of features with 5-fold cross validation. As it can be seen in table 1, the KNN networks trained using the training set have produced exemplary for the features extracted frame-by-frame. A total of 206858 observations were utilized by the network, producing highest results for the joint location features. Despite being challenging to extract accurate 2D skeletons using OpenPose, the networks have trained with fairly high accuracy in a minimal training time.

Table 1: Comparison of network results produced from all feature sets

Dataset	Methodology	Violence(Class 1)	Non-violence (Class 2)	Classifier	Training time	Overall accuracy
RWF2000	**Joint location for arms**	**99.4%**	**96.4%**	**Fine KNN**	**5.8625s**	**98.6%**
	Joint angles for shoulder and elbow	97.5%	82.2%	Fine KNN	12.744s	93.5%
	Temporal patches features	99%	91.3%	Fine KNN	5.4109s	97.0%

In table 2, the results produced from the trained networks using the testing set are shown. In this case, the accuracy is much lesser than the benchmark results. The reason being that some of non-violent videos do not contain people rather just cars or other scenes. In such a scenario the number of frames containing 2D skeletons is decreased by even less than half, hence resulting in lower accuracy.

Table 2: Comparison to benchmarks on the testing set

Dataset	Reference	Methodology	Classifier	Accuracy
RWF2000	Ours	Joint location for arms features	Fine KNN	71.2%
		Joint angles for shoulder and elbow	Fine KNN	65.5%
		Temporal patches features	Fine KNN	67.0%
	[43]	Flow gated network	Deep learning	85.75%

In this paper, an approach for detection violence using affect detection models was proposed. A set of low-level features was computed using the 2D skeleton data extracted through OpenPose toolbox. The features were then used to train Fine KNN classifiers yielding promising results in low training time. However, the results produced using the testing set was not upto the benchmark. As the features could not be extracted for many non-violent videos due to the absence of people. In future, features such as Quantity of Motion (QoM), Motion History Image (MHI) and Optical flow will be incorporated along with the affective anger features in order to produce more accurate results. Also, other datasets will be considered for thoroughly analysing the efficiency of the proposed features.

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